SYNX THERAPEUTICS · MATS 11.0

Pre-registration: does the coverage mechanism hold on independent proteins?

Written 20 October 2026, before the replays were run. Commit this file first; the result goes in a separate commit. The point of writing it down is that a prediction recorded after the fact is not a prediction, and this project has already retracted three claims in two days.

What is being tested

The surviving mechanism from *ADJUDICATION_COMBINATORIAL_GAIN.md* is that acquisition-policy gain is governed by whether the shared surrogate can learn **inside the candidate pool**. Its top-ranked unresolved confound is independence: the combinatorial arm rested on one independent protein (YAP1 WW domain) plus a restaging of the pilot protein (GB1), so the landscape-level permutation floor sat at $p = 1/15 = 0.067$ and could not fall.

This adds two landscapes from unrelated proteins at the same 512/128/5% geometry:

LANDSCAPE	PROTEIN	VARIANTS	MULTI-SUB	ENDPOINT	VARIABLE POSITIONS
`grb2_sh3_peptide`	GRB2 SH3 domain	63,366	62,332	binding-coupled yeast growth	56
`rask_darpin_binding`	KRAS vs DARPin K55	24,873	21,789	binding, by yeast growth	187

Neither is related to GB1, YAP1 or any landscape already staged. `rask_darpin_binding` also carries a binding rather than an organismal-fitness endpoint, which partly breaks the assay-class confound in the earlier six.

The predictor, measured before the replays

In-pool surrogate skill: fit the project's own 64-tree forest on 128 randomly chosen labels from a 512-candidate pool, then take Spearman against truth on the 384 held out. Averaged over the first five pools. This is the mechanism stated as a measurable quantity.

LANDSCAPE	VARIABLE POSITIONS	IN-POOL SPEARMAN	GREEDY GAIN (ALREADY KNOWN)
`spg1_olson_igg_matched`	55	0.695	+0.242
`rask_darpin_binding`	**187**	**0.675**	**— to be measured**
`grb2_sh3_peptide`	**56**	**0.455**	**— to be measured**
`yap1_ww_peptide`	34	0.345	+0.263
`ha_h1`	2	0.119	−0.004
`spike_rbd_ace2`	2	0.112	+0.030
`mpro`	2	0.091	+0.065
`ha_h3`	2	0.075	+0.060 (all 20 pools)

On the six already measured, the split is clean: in-pool Spearman at or above 0.34 goes with a gain at or above +0.24; at or below 0.12 with a gain at or below +0.065. Nothing yet lies between 0.12 and 0.34, which is itself a gap worth filling later.

The prediction

Both new landscapes sit above the threshold, so the mechanism predicts a substantial gain in both.

1. **Primary, falsifiable.** Greedy gain over random at 128 labels, zero delay, paired per pool, 10 disjoint pools: **≥ +0.15 for both landscapes**, improving in **≥ 7 of 10 pools** each.
2. **Point predictions.** `rask_darpin_binding` **+0.24 ± 0.10**; `grb2_sh3_peptide` **+0.25 ± 0.10**. I am deliberately not ordering them by Spearman, because gain was already shown not to be monotone in surrogate skill — GB1-matched at 0.695 gives +0.242 while YAP1 at 0.345 gives +0.263.
3. **UCB** behaves similarly to greedy; **diverse** shows no gain (within ±0.05 of zero) on both, since the effect is a surrogate effect and diverse uses no surrogate.

What would falsify it

- Either landscape showing greedy gain **below +0.10** falsifies the mechanism as stated. It would mean in-pool learnability is not sufficient even as a tendency, and the claim would have to retreat to a description of the six original landscapes.
- Either showing a gain with **diverse** comparable to greedy would mean the effect is not a surrogate effect, and the mechanism is misidentified.
- Gains in the +0.10 to +0.15 band would be an equivocal result and will be reported as such, not rounded into support.

What this test does *not* do

It does **not** separate the coverage mechanism from the discarded "combined substitutions" explanation: both landscapes are multi-substitution and high-coverage, so both hypotheses predict the same outcome here. This test addresses independence only — whether the effect exists on proteins unrelated to the two it was found on. Separating coverage from combinatoriality needs the within-protein 2×2 (PDZ doubles against the already-staged PDZ singles with coverage matched), which is a separate experiment and is not run here.

It also does not bear on macrocycles, display readouts, or any pool larger than 512; the gain is already known to decay steeply with pool size at fixed budget.

Analysis, fixed in advance

Ten disjoint evaluation pools per landscape, policies random / diverse / greedy / ucb, zero delay, 128 labels, top-5% recall, paired per pool against random on the same pool with the same eight seeded initial labels. Report mean paired gain, standard deviation across pools, and the count of pools improved. No multiple over random will be quoted. If the exact landscape-level permutation test is run, it will be reported with its floor (4 versus 4 gives $1/70 = 0.014$).

Outcome, appended 20 October after the replays were run

Everything above this line was committed before the replays (`3f9b0a1`-series, see `git log --follow PREREGISTRATION_COVERAGE.md`). Nothing above has been edited.

All criteria passed. At 10 pools, the geometry the prediction was written for:

LANDSCAPE	PREDICTED	OBSERVED GREEDY	POOLS IMPROVED	DIVERSE
`grb2_sh3_peptide`	+0.25 ± 0.10, ≥ 7/10	**+0.254**	9/10	−0.035
`rask_darpin_binding`	+0.24 ± 0.10, ≥ 7/10	**+0.254**	8/10	+0.038

Primary criterion (≥ +0.15 and ≥ 7/10): pass on both. Point predictions: within 0.005 on both, which is closer than the method deserves and should not be read as precision. Diverse within ±0.05 of zero: pass on both, consistent with a surrogate effect.

On the remaining ten pools, run afterwards. Both landscapes were staged with 20 pools; the prediction was written for 10, so the second half is an out-of-sample extension rather than part of the registered test. GRB2 strengthened to +0.265 over 20 pools (19/20). KRAS weakened to +0.194 (15/20) — outside the registered ±0.10 band had the test been written for 20 pools. Both figures are reported, and the 20-pool numbers are the ones used everywhere else, because using the half that was registered while twenty exist is the selective reporting this file is meant to prevent.

Effect on the landscape-level comparison. Four high-coverage landscapes against four low-coverage, all staged pools run: high-coverage +0.194 to +0.265, low-coverage −0.004 to +0.065, no overlap. Exact one-sided permutation $p = 1/70 = 0.014$, the floor of the design, against $1/15 = 0.067$ before these two were added.

What it does not establish, restated because it has not changed: this tests independence, not the coverage-versus-combinatoriality distinction, and bears on neither macrocycles nor display readouts nor pools larger than 512.